Do Now · retrieval

Programming Techniques · Functions and Procedures

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the final result of x=5; x-=2;?

Q2

What does a binary search need first?

Q3

How many times runs for(i=0;i<4;i++)?

Q4

What does a truth table list?

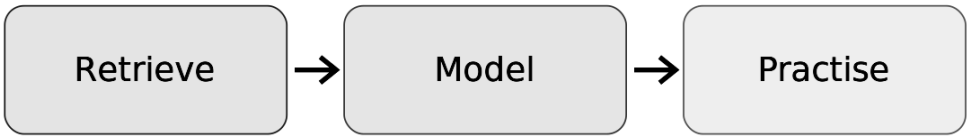
New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Explain the important decision — The argument is copied into a local parameter. return sends a computed value to the caller; it does not itself print the value.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

function — Returns a result

procedure — Subroutine that does not return a result

parameter — Named input to a subroutine

argument — Value passed in a call

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Return a computed value

Follow the worked method

1. Pass the argument — Double(6) starts a call with local parameter value=6. The caller waits for an integer result.

2. Calculate and return — value*2 evaluates to 12. return 12 ends this call and supplies that integer to the caller.

3. Use the returned value — The assignment stores 12 in result; Console.WriteLine(result) prints it. A void procedure can print, but supplies no value for this assignment.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Discuss, then respond
A function adds a and b; what is Add(3, 4)?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Apply independently

Write a C# method returning the cube of an integer. [4]

Question 2

Apply independently

Explain the purpose of a parameter. [2]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Apply independently

Double(n) returns n*2. Calculate Double(Double(2)) and show both returns. [2]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

True or false: a void C# method must return a calculated value.

Use key vocabulary from memory

1. Explain function in your own words.

2. Explain procedure in your own words.

3. Explain parameter in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.